Pranav Senthilkumar

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Education

Texas A&M University — College Station, TX
Bachelor of Engineering, Engineering Honors — *Aug. 2025 – May 2029*

Experience

Student Researcher — December 2025-Present
Texas A&M University — *College Station, TX*

- Architecting a sonar-based sensing system for subsurface evaluation of bridge foundation stability (TxDOT collaboration), focusing on signal acquisition, processing, and geospatial integration
- Developing a scalable pipeline for processing echo sounder and side-scan sonar data, including signal conditioning, feature extraction, and visualization of underwater terrain
- Exploring machine learning methods for anomaly detection in sonar signals to identify subsurface movement and geotechnical instability

Research Team Lead — June 2024 – October 2024
Algoverse AI Research — *Remote*

- Conducted applied research on calibration and alignment of large language models in ambiguous real-world decision contexts
- Fine-tuned transformer models (Llama-3, Mistral-7B, Zephyr-7B) using QLoRA, achieving up to 31% improvement in Dirichlet loss
- Designed evaluation pipelines using ethical reasoning datasets to benchmark probabilistic alignment and model behavior under uncertainty

Skills

Programming Languages: Python, C++, Java
Scientific / Data Libraries: NumPy, Pandas, OpenCV, MediaPipe
AI / ML: TensorFlow, PyTorch, Keras, SciKit Learn
Systems & Tools: GitHub, Firebase, Arduino, KiCad (learning), Microsoft Flight Simulator

Projects

Haptic Portal — January 2026 – Present
Python, DepthAI, OpenCV, MediaPipe, UDP, KiCad

- Developing a real-time vision-to-haptics system that converts hand tracking and Oak-D-Lite camera stereo depth sensing into tactile control signals for interactive haptic devices
- Implemented a depth-processing module that converts stereo depth maps into a compact 5×5 feature matrix, reducing high-resolution depth data into 25 normalized channels for real-time control
- Added temporal filtering and median smoothing to stabilize hand-depth measurements across frames, improving reliability of the real-time haptic signal
- Designing a PCB integrating a Raspberry Pi Pico and PCA9685 drivers, with a structured layout and managed power routing for reliable operation

MCU Data Logger PCB — December 2025 – January 2026
KiCad, Git

- Designed a 4-layer MCU-based data logger PCB using an ATmega328P-AU, I^2C RTC (DS1337), and 512 Kb external EEPROM to support persistent, timestamped data storage
- Engineered a I^2C and power architecture with dedicated ground/power planes, proper decoupling, pull-ups, and timing-critical crystal/RTC layout
- Managed the complete hardware design lifecycle in KiCad using Git for version control, producing DRC/ERC-clean schematics, PCB layout, BOM, and fabrication-ready outputs

Publications

- Pranav Senthilkumar, et al. *Fine-Tuning Language Models for Ethical Ambiguity: A Comparative Study of Alignment with Human Responses*. Accepted at NeurIPS 2024, SoLaR Workshop. arXiv:2410.07826